Community Detection in Georgian News Networks: A Consensus Framework

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1 Introduction

Understanding the political inclinations of news sources can be helpful to interpret news in a more objective way by taking political bias into account. However, as an outside observer, it is not always easy to analyze the bias that colors news stories: this requires extensive experience in reading from diverse sources or relying on subjective assessments of domain experts. To address this, previous studies use network structure successfully to detect political bias for networks consisting of political blogs [1] and twitter users [7].

In a study more relevant to our own, Lin et al. [12] construct networks using hyperlink structure and content similarity of a wide range of online news outlets. Using community detection, strong ideological structure emerges for both networks. Particularly the hyperlink network shows strong ideological clusterings, where news outlets with similar political bias tend to cite each other more often. The content similarity network also displays ideological structuring, although somewhat less pronounced. These findings promote the idea that political bias can be derived from structural patterns in a network of news outlets.

Similar to their work, our goal is to analyze and detect clusters that may represent politically aligned media sources in a network of Georgian media outlets by applying community detection algorithms. These networks are created by assigning news sources to topic-based clusters on a per-day basis and aggregating these over multiple days, where edges between similar media sources are stronger.

However, since no definitive ground truth exists for the alignment of political bias of Georgian news outlets, we do not aim to validate political inclination directly. Instead we pose the following research question: Does a consensus clustering emerge, given different clusterings methods of network construction and community detection?

To answer this question, we evaluate the degree of agreement among a wide range of community detection results, each generated using a wide combination of different methods of network creation, community detection and a corresponding set of hyper-parameters. This is in line with Ghasemian et al. [8], who argue that a single model can overfit or underfit on the data, stressing the importance to test multiple configurations of community detection methods.

This paper presents an experimentation framework to detect a consensus of communities emerging from community structures formed by comparing an extensive range of methods of network creation and community detection.

We hypothesize that news sources appear in the same communities consistently across different configurations. This is based on the theory that news sources that are politically aligned publish articles concerning the same topics with relatively higher frequency, which creates stronger network links, leading to more consistent community assignments.[1]

2 Data Description

We investigate our hypothesis with a Georgian news dataset containing practically all online news coverage from the country over a

[1]Note that if our hypothesis is correct, we can not confirm whether they are in fact politically aligned, merely that we indeed found that there are indeed consistent communities.

period of 8 months, with a total of approximately 240,000 articles from 49 media sources in this period. For each day, we semantically cluster the news articles into groups covering the same stories. We then form a square co-occurrence matrix N, where N_{ij} is the number of times source i and j were in the same cluster across the 240 days. For each day, a cluster roughly corresponds to a news topic. Although this already constitutes an adjacency matrix, several modeling choices need to be made to account for significant variability in the prolificacy of different news sources and other domain-specific considerations.

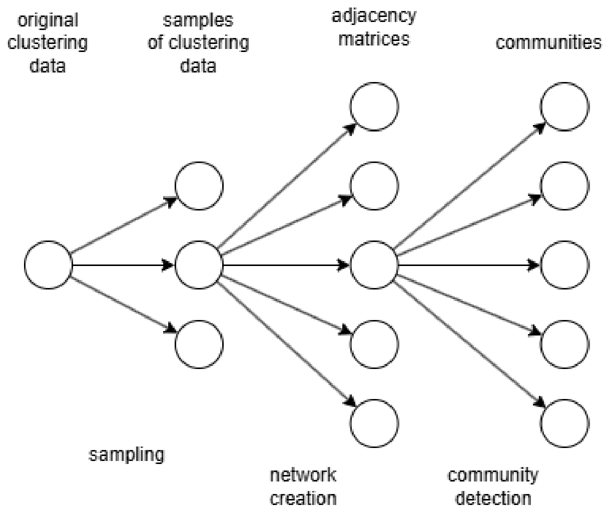

Figure 1: Diagram of the experimental framework

3 Methodology

3.1 Evaluation Metrics

To assess whether a given pair of communities is similar, we use the adjusted Rand index (ARI) [10], which is a measure of the similarity of two clusterings, while taking into account the chance of random groupings occurring. ARI works by comparing all pairs of elements and counting pairs that are assigned together or apart in two given clusterings. It then takes into account the number of agreements that would be expected by chance. The resulting metric can range from 1 to -1. An ARI score of 1 indicates perfectly identical clusterings, a score of 0 indicates that the clusterings is what would be expected from random chance, and an ARI of less than 0 would indicate that the clusterings are less similar than what would be expected from random chance.

3.2 Data sampling

We begin the experiment by sampling sixteen 15-day intervals from the clustered news data. The samples are consecutive and without overlap, and together cover 240 days of Georgian news coverage across a total of n=49 sources.

3.3 Network Modeling and Community Detection

After sampling the data, we transform each sample into network adjacency matrices using the network modeling approaches outlined in appendix tables A.1 and A.2. This results in directed and undirected graphs with weighted edges representing connection strength between two media outlets, represented by adjacency matrices.

Different network modeling strategies reflect distinct assumptions about what constitutes a meaningful connection. Frequency-based methods, such as raw co-occurrence counts, provide a straightforward and intuitive view of which outlets appear together most often. However, they are susceptible to *prolificity bias*: outlets that publish frequently or are covered across many clusters tend to dominate the network regardless of their true thematic alignment. This can lead to inflated edge weights and misleading community structures driven more by publication volume than actual underlying similarity.

This is why we include normalized similarity measures such as Jaccard and Dice coefficients. These down-weight the influence of prolific outlets by focusing on relative overlap rather than absolute counts. Similarly, vector-based methods like cosine similarity and TF-IDF help adjust for differences in how often outlets occur, giving a more focused view of how they really relate to each other. Probabilistic measures such as Pointwise Mutual Information (PMI) and Lift further highlight strong associations, which might otherwise be obscured in raw frequency data. Conditional probability helps capture asymmetric or directional relationships, useful when one outlet consistently co-occurs with another, but not vice versa.

By incorporating this diverse set of network modeling techniques, we aim to explore community structures that are robust to different underlying assumptions about similarity and co-occurrence. This diversity is essential for reducing bias and capturing multiple, potentially overlapping forms of media alignment.

Once networks are constructed, we apply a variety of community detection algorithms to identify clusters of closely connected outlets. As detailed in table A.1, these include modularity-based methods such as Louvain and Leiden, information-theoretic approaches like Infomap, label propagation, and spectral clustering. Each algorithm captures a different intuition about what constitutes a "community" from local edge density to random walk behavior or network flows.

We also explore simpler, threshold-based approaches such as k-core decomposition and connected components after edge pruning, which provide more interpretable, rule-based segmentations of the network. Where supported, we tune hyperparameters such as resolution (for modularity-based methods), number of clusters (for spectral methods), and thresholds (for pruning-based techniques) to evaluate how algorithmic and parametric choices influence detected community structure.

3.4 Aggregation

At the end of the above process, we have 8336 clusterings resulting from 16 time windows × network modeling methods × (9 community detection algorithms × their hyperparameter instances). In order to ultimately analyze significant pairwise relationships between sources and derive a robust final clustering, we aggregate the results within each sample.

Given that each run results in varying numbers of communities of different sizes, it is crucial to select an appropriate result aggregation process. Using a naive approach, we can derive an outlet co-clustering frequency matrix by counting how often different outlets were clustered together across all runs. However, this introduces a subtle bias into our analysis. Two outlets clustering together in a trivial community with $n-2$ other outlets contributes no information to our understanding of their affinity for one-another; to a lesser extent, this consideration extends to large non-trivial communities. On the other hand, two sources often clustering as part of smaller communities is more indicative of their likeness.

3.5 Information-Weighted Pairwise Affinity

We address this limitation by computing the pairwise co-clustering affinities using an information-theoretic surprisal weighting scheme that accounts for cluster size bias. The surprisal for a pair of outlets clustered together in a community of size $||c||$ is defined as:

$$s = -\log_2\left(\frac{||c||}{n}\right)$$

where n is the total number of outlets. Given an observation of a pair of sources clustered together, this weighting scheme assigns a weight to the observation which is equivalent to its information content, in bits, which is inversely proportional to the probability of their clustering together by chance given the size of the cluster. As a result, the weighting scheme up-weights the affinity between pairs when they appear in smaller clusters, while down-weighting pairs that co-occur in large clusters that may arise by chance. The source-pair affinity measure, which we refer to as the information-weighted pairwise affinity (IPA) between outlets i and j, is then computed as:

$$A_{ij} = \frac{1}{K}\sum_{k=1}^{K} s_{ij}^{(k)} \cdot \mathbf{1}[c_i^{(k)} = c_j^{(k)}]$$

where K is the total number of clustering results, $c_i^{(k)}$ represents the cluster assignment of outlet i in clustering result k, and the surprisal weight is applied only when outlets are co-clustered:

$$s_{ij}^{(k)} = \begin{cases} -\log_2\left(\frac{||c^{(k)}||}{n}\right) & \text{if } c_i^{(k)} = c_j^{(k)} \\ 0 & \text{if } c_i^{(k)} \neq c_j^{(k)} \end{cases}$$

We note that this surprisal-based weighting scheme has some intuitive properties. At clus-

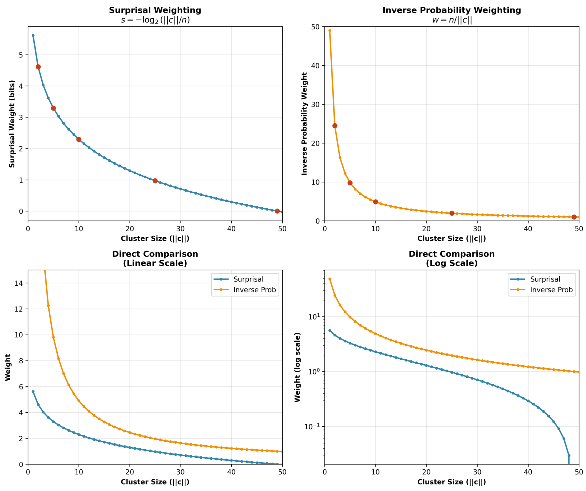

Figure 2: Comparison of weighting schemes for cluster size bias correction

ter size $||c|| = n$, the resulting surprisal is exactly $s = -\log_2(n/n) = -\log_2(1) = 0$, corresponding to 0 bits of information – an uninformative observation. At $||c|| = n/2$, the surprisal is exactly $s = -\log_2((n/2)/n) = -\log_2(1/2) = 1$ bit, as befits a 50% chance of the observation occurring randomly. Surprisal is also additive, as bits of information can be aggregated by taking their sum. This fits our analysis, which relies on summing the surprisal across clusterings.

Exploring alternatives to surprisal-based weighting, we might consider weighting by the inverse of the probability of falling within a cluster of size $||c||$:

$$w = \frac{1}{P} = \frac{1}{||c||/n} = \frac{n}{||c||},$$

which, like surprisal, down-weights the significance of pairs in larger clusters. This scheme, however, results in extreme values at small cluster sizes. We can apply a log-transformation to the inverse probability, which is simply equivalent to the surprisal measure we've already introduced. Alternatively, we might opt to scale linearly by the complement of the probability, $w = 1 - P$. However, this measure fails to meaningfully differentiate pair affinity between comparable cluster sizes in ranges where such differences particularly matter (figure 2). With this in mind, we believe our use of surprisal is well-justified.

3.6 Statistical Significance Testing (cross-sample approach)

To establish the statistical significance of pairwise co-clustering relationships, we employ a cross-sample testing framework that leverages the temporal independence of our 15-day windows. For each of the S temporal windows, we aggregate the K_s clustering runs within that window into a single surprisal-weighted co-clustering matrix (as described in an earlier section), yielding S independent information-weighted pairwise affinity (IPA) matrices, one per temporal sample.

Sample-specific null model. For each sample s, we compute an analytical null distribution based on the specific cluster size distributions observed in that window. For a single clustering result with cluster sizes $\{s_1, \ldots, s_c\}$, the expected surprisal-weighted frequency is:

$$\mu_k = \sum_{\ell=1}^{c} p_\ell w_\ell, \qquad \sigma_k^2 = \sum_{\ell=1}^{c} p_\ell w_\ell^2 - \mu_k^2,$$

where $p_\ell = \frac{s_\ell(s_\ell - 1)}{n(n-1)}$ is the probability that a random pair falls within cluster ℓ, and $w_\ell = -\log_2(s_\ell/n)$ is the corresponding surprisal weight. Averaging across the K_s runs yields sample-level null parameters $\mu_{ij}^{(s)}$ and $\sigma_{ij}^{2\,(s)} = K_s^{-2} \sum_k \sigma_k^2$.

Cross-sample residual analysis. We stack the S samples into tensors $X \in \mathbb{R}^{S \times n \times n}$ (observed IPAs) and M (null expectations), then compute residuals $Y_{s,ij} = X_{s,ij} - M_{s,ij}$ for each outlet pair across samples. This residual series captures how much each pair's co-clustering deviates from chance expectations in each temporal window.

Outlet pairs are deemed testable if they exhibit positive residual variance in at least $\lceil S/2 \rceil$ samples. For testable pairs, we apply the two-sided Wilcoxon signed-rank test to the residual series, which tests whether the median residual differs significantly from zero without assuming normality.

Multiple-testing correction and classification. All p-values undergo Benjamini–Hochberg FDR correction ($\alpha = 0.05$)

to control the false discovery rate across the multiple pairwise comparisons. Significant pairs are classified as either *high* co-clustering ($\bar{A}_{ij} > \mu_{ij}^{\text{null}}$) or *low* co-clustering ($\bar{A}_{ij} < \mu_{ij}^{\text{null}}$), indicating systematic tendencies to cluster together more or less frequently than expected by chance.

3.7 Final Clustering from Validated Affinities

The statistically validated relationships form the foundation for our final consensus clustering. We construct a signed similarity matrix that preserves only statistically significant relationships:

$$S_{ij} = \begin{cases} \bar{A}_{ij}, & \text{if pair } (i,j) \text{ is significant,} \\ \mu_{ij}^{\text{null}}, & \text{otherwise.} \end{cases}$$

With $S_{ii} = 1$ to ensure self-similarity on the diagonal.

Thus, significant high-affinity pairs retain their observed values (above null expectation), significant low-affinity pairs retain their observed values (below null expectation), and non-significant pairs are set to the null expectation. This creates a signed similarity matrix where values above/below the null baseline provide positive/negative evidence for clustering relationships.

We convert the similarity matrix to a distance matrix via $D_{ij} = S_{\max} - S_{ij}$ where $S_{\max} = \max_{i,j} S_{ij}$, such that high affinities correspond to small distances. We then apply Ward's minimum-variance linkage to the condensed distance matrix, producing a hierarchical clustering that minimizes within-cluster variance at each merge step. The resulting dendrogram captures a hierarchy of consensus communities. We determine the final partition using the elbow method.

4 Results

4.1 Experimental Overview and Data Characteristics

Across n = 49 news outlets, we ran 8,336 distinct community-detection configurations (16 temporal windows × 10 network-modelling methods × 9 community algorithms × hyperparameter grids). After filtering trivial partitions $k = 1$ or $k = n = 49$, we retained 6,191 results - 74.3 % of the full grid. Exclusion results are displayed in Figure A.4. The distribution of clustering modularities is summarized by Figure A.5 (mean: 0.102, std: 0.130). The relationship between modularity and k is summarized in Figure A.6 (k-modularity correlation = 0.202).

4.2 Method Stability

4.3 Information-Weighted Affinity

Aggregating all retained clusterings with surprisal weights produced a mean IPA matrix A (Figure A.7) (mean = 0.2857, std = 0.2027, max = 1.3145, compared to the raw frequency matrix mean: 0.4061, std: 0.1342, max: 0.8230). Difference between the value of the raw and surprisal-weighted matrices (weighted - raw) centered at -0.1204 with std 0.1115 and range [-0.2500, 0.5070]. A comparison between the two matrices can be found in Figure A.7.

4.4 Significance Testing

For each of the 16 independent temporal windows we derived an IPA matrix and compared it against an analytical null expectation (Section 3.5). Source pairs were evaluated with a two-sided Wilcoxon signed-rank test and Benjamini–Hochberg correction ($\alpha = 0.05$). We find 251 pairs with significantly high co-clustering and 594 with significantly low co-clustering. Figure A.10 visualises residual effect sizes (observed – expected).

4.5 Consensus Communities

Replacing non-significant pairs with their null expectation yielded a signed similarity matrix S. Ward linkage on the derived distance matrix produced the hierarchical consensus shown in Figure A.12; the elbow criterion suggested $k = 2$ clusters, with a median within-community IPA: 0.571 (IQR 0.395–0.707), median between-community IPA: 0.126, and a signed modularity Q: 0.343. Figure A.12 illustrates the hierarchical clustering. Figure 3 depicts the graph of statistically significant edges.

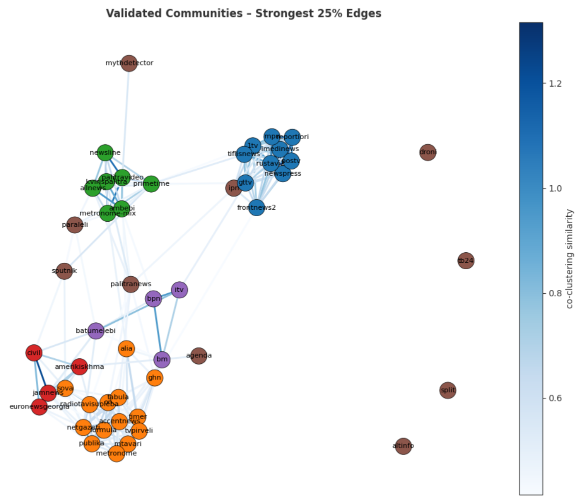

Figure 3: Resulting communities

4.6 Temporal Robustness

Pairwise ARI between consecutive 15-day windows averages 0.360 (std. 0.109). Figure A.8 depicts the mean ARI between the 15-day samples, and Figure A.9 shows the time-lag similarity profile between different windows.

5 Discussion and Conclusion

While it's impossible to compare the results to a ground truth, the resulting clusterings are intuitively extremely appropriate, given the domain knowledge of one author. For instance, the blue cluster corresponds to the most pro-government outlets, while the orange cluster corresponds to the pro-opposition ones.

Our experimental framework demonstrates that a consensus clustering emerges when applying diverse community detection methods to Georgian news outlet networks. Across 8,336 distinct configurations, we consistently observe structured community patterns that persist beyond what would be expected from random chance, providing strong evidence for our central research question.

The surprisal-weighted aggregation scheme down-weights trivial large-cluster associations while emphasizing meaningful small-cluster relationships, as evidenced by the substantial differences between raw and weighted affinity matrices (mean difference: -0.1204, std: 0.1115). We are most proud of this contribution.

The network modeling approach, while comprehensive, relies on topic-based clustering as a proxy for content similarity. This intermediate step may introduce noise if the semantic clustering fails to capture subtle ideological differences in coverage of the same events. Additionally, our framework cannot distinguish between editorial alignment and other factors that might drive co-clustering, such as shared news sources, syndication agreements, or market positioning strategies.

The methodological diversity, while a strength for robustness, also introduces complexity in interpretation. Different network construction methods and community detection algorithms capture distinct aspects of media relationships, and the aggregation process necessarily involves information loss. The choice of hyperparameter grids, while extensive, may not capture all relevant parameter spaces for each algorithm.

Despite these limitations, the emergence of consistent community structure across such a diverse methodological landscape provides compelling evidence for systematic patterns in Georgian media outlet relationships. The framework demonstrates that consensus can emerge from methodological diversity, supporting the hypothesis that structurally aligned news sources exhibit consistent co-clustering behavior regardless of specific analytical choices.

6 AI Statement

We used AI to write and refactor code and aid with the writing of this paper. However, we used AI only to augment, rather than replace our own work. The resulting analysis is far more comprehensive than we could've achieved otherwise.

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A Appendix

Name	Category	Description	Strengths / Limitations
Co-occurrence [15]	Frequency-based	Counts the number of times two sources appear together within the same cluster or context, providing a straightforward measure of association based on raw occurrence data.	Simple, intuitive; biased by prolific sources
Jaccard Similarity [11]	Set-theoretic / Statistical	Measures the overlap between two sets divided by their union, quantifying the similarity of co-cluster appearances by normalizing for their total combined presence.	Normalized; sensitive to sparse overlap
Dice Coefficient [6]	Set-theoretic / Statistical	Calculates similarity by doubling the intersection size over the sum of the sizes of both sets, emphasizing mutual presence rather than just overlap.	Emphasizes mutual presence; similar to Jaccard
Cosine Similarity [20]	Vector-based	Computes the cosine of the angle between raw co-cluster frequency vectors, capturing similarity based on vector orientation regardless of magnitude differences.	Handles magnitude variation; assumes continuous weights
Pearson Correlation [2]	Statistical	Evaluates the linear correlation between co-cluster vectors, measuring how changes in one variable predict changes in another.	Captures linear relationships; needs normalization
Pointwise Mutual Info [4]	Probabilistic / Information-theoretic	Measures how much more often two items co-occur than expected by chance, highlighting statistically significant associations between sources.	Highlights strong co-occurrence; sparse for rare pairs
Lift [23]	Probabilistic / Statistical	Quantifies the strength of association rules by comparing observed co-occurrence probability to the expected probability if sources were independent.	Sensitive to rare events; can be noisy
TF-IDF Similarity [22]	Vector-based / Frequency-weighted	Applies term frequency-inverse document frequency weighting before computing cosine similarity, reducing the influence of common clusters to better highlight distinctive links.	Downweights common clusters; good for contrastive links
Conditional Probability [13]	Probabilistic	Computes the probability of occurrence of one source given the presence of another, capturing directional and potentially asymmetric relationships between items.	Captures directional relationships ($P(j\|i)$)

Table A.1: Comparison of Network Creation methods

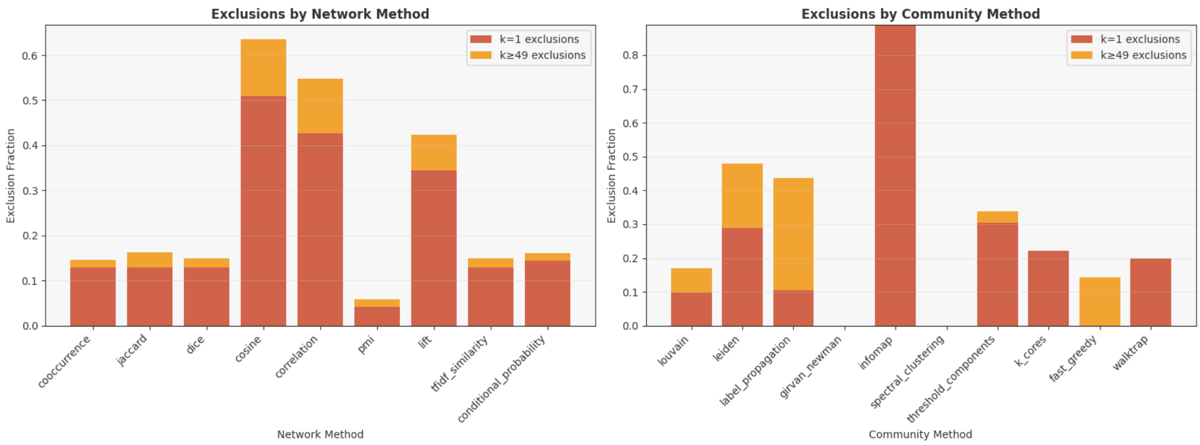

Figure A.4: Exclusion statistics by method

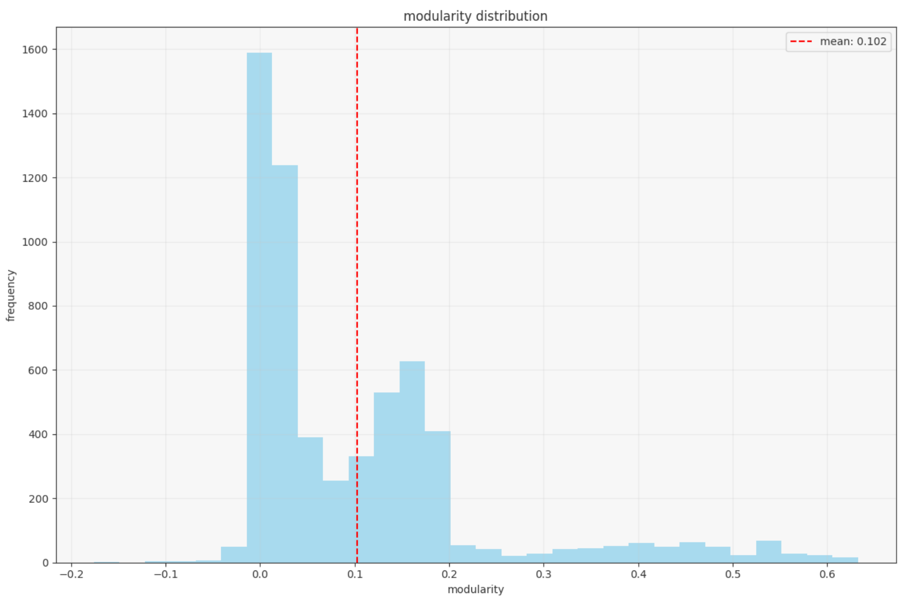

Figure A.5: Distribution of clustering modularities across all methods

Name	Description	Parameters	Library	Strengths / Limitations
Louvain [3]	Iteratively moves nodes between communities to maximize modularity gain, using local optimization followed by community aggregation	resolution: [0.1, 0.3, 0.5, 0.8, 1.0, 1.2, 1.5, 2.0, 2.5]	community	Fast and scalable; resolution limit
Leiden [24]	Refines Louvain by first moving nodes to singleton communities, then merging communities, ensuring well-connected communities	resolution: [0.01, 0.05, 0.1, 0.2]; objective_function: ['modularity', 'CPM']	leidenalg	Improved over Louvain; supports disconnected graphs
Label Propagation [18]	Each node iteratively adopts the most frequent label among its neighbors until convergence	None	NetworkX	Very fast; non-deterministic; no control over number of communities
Girvan-Newman [9]	Recursively removes edges with highest betweenness centrality to reveal community structure through hierarchical splitting	None	NetworkX	Clear structure; not scalable to large graphs
Infomap [19]	Maps flow of random walks on networks, partitioning to minimize description length of trajectories	num_trials: [1, 5, 10]; two_level: [True, False]	Infomap	Good for directed/flow structures; sensitive to noise
Spectral Clustering [16]	Computes eigenvectors of graph Laplacian, embeds nodes in low-dimensional space, then applies k-means clustering	n_clusters: [2,3,4,5,6,8]; eigen_solver: ['arpack','lobpcg'] assign_labels: ['kmeans','discreti	sklearn	Assumes known # of clusters; works well for dense graphs
Threshold Components [14]	Removes edges below specified weight threshold, then finds connected components in remaining graph	threshold: [0.01, 0.02, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5]	NetworkX	Very interpretable; sensitive to threshold choice
k-Cores [21]	Iteratively removes nodes with degree less than k until all remaining nodes have degree $\geq$ k	k: [1, 2, 3, 4, 5]	NetworkX	Simple structural insight; may oversimplify
Fast Greedy [5]	Builds dendrogram by greedily merging communities that maximize modularity increase at each step	weights: [True, False]	iGraph	Good for hierarchical data; can be slow
Walktrap [17]	Computes distances based on short random walks, then applies hierarchical clustering on walk-based distances	steps: [2,3,4,5,6,8]; weights: [True, False]	iGraph	Good balance of speed and accuracy; may over-merge communities

Table A.2: Comparison of Community Detection Algorithms

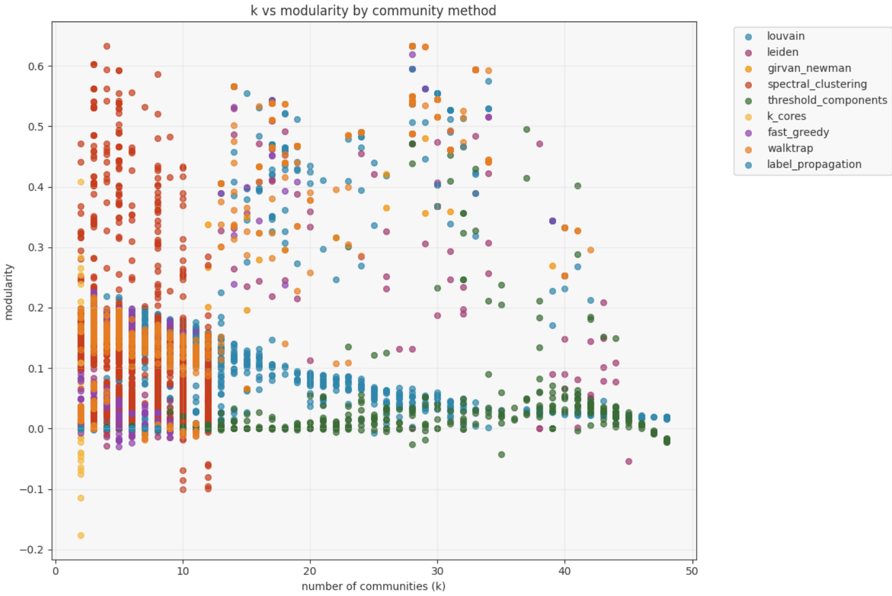

Figure A.6: K vs Modularity by community detection method

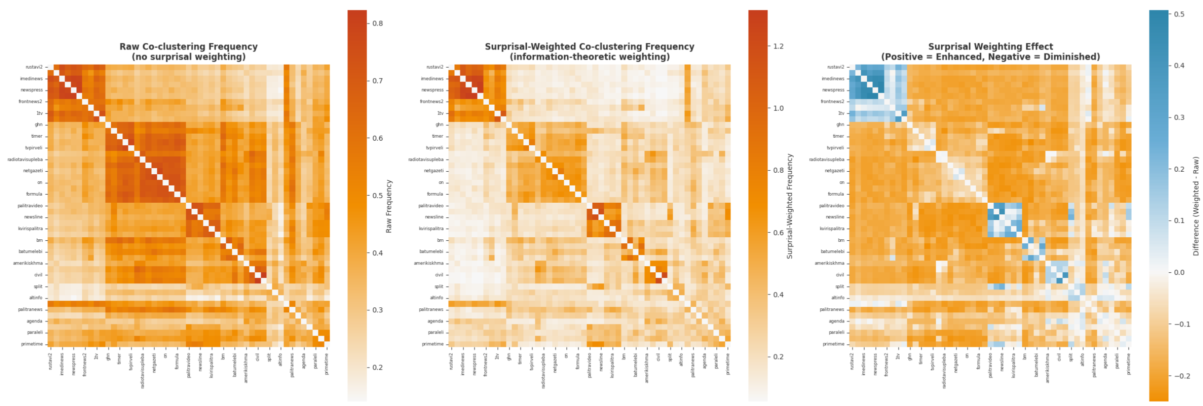

Figure A.7: Affinity matrices: surprisal-weighted vs raw comparison.

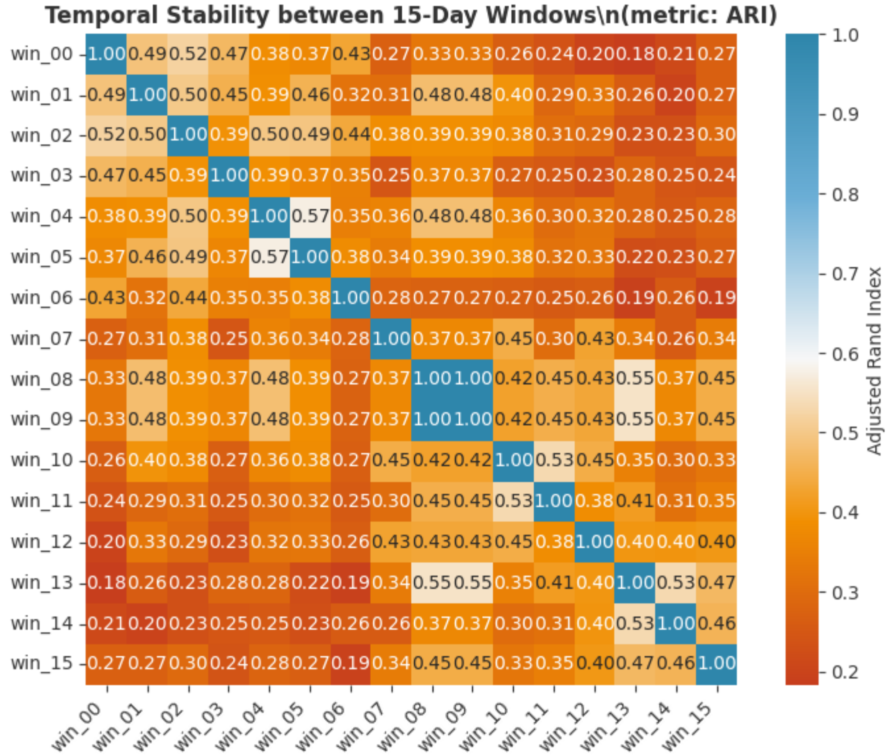

Figure A.8: Temporal stability between time samples

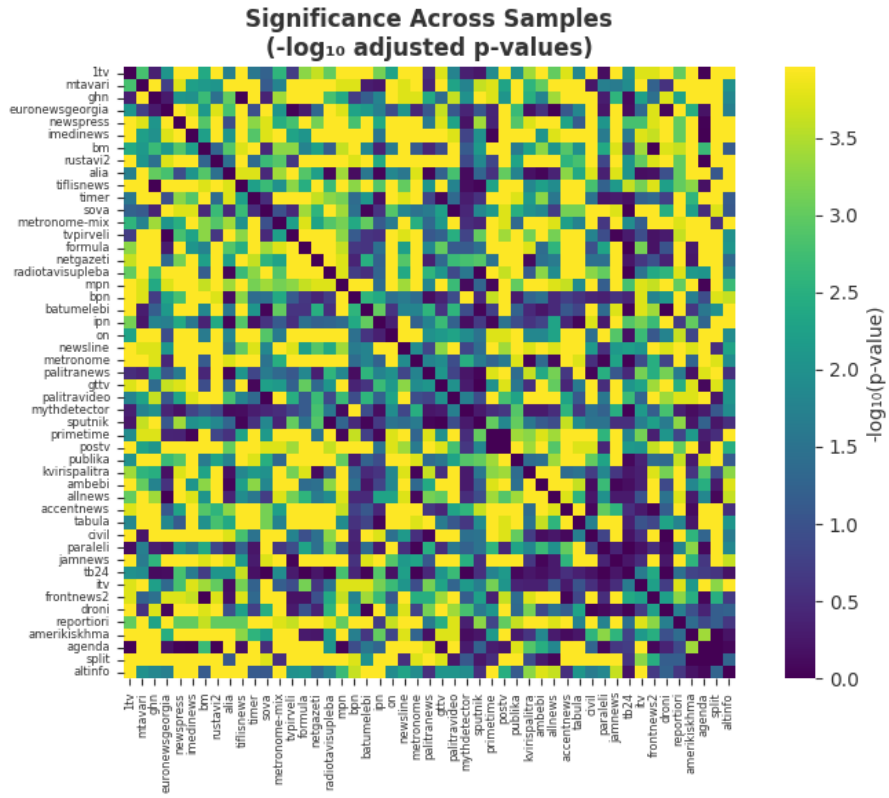

Figure A.9: Significance across samples

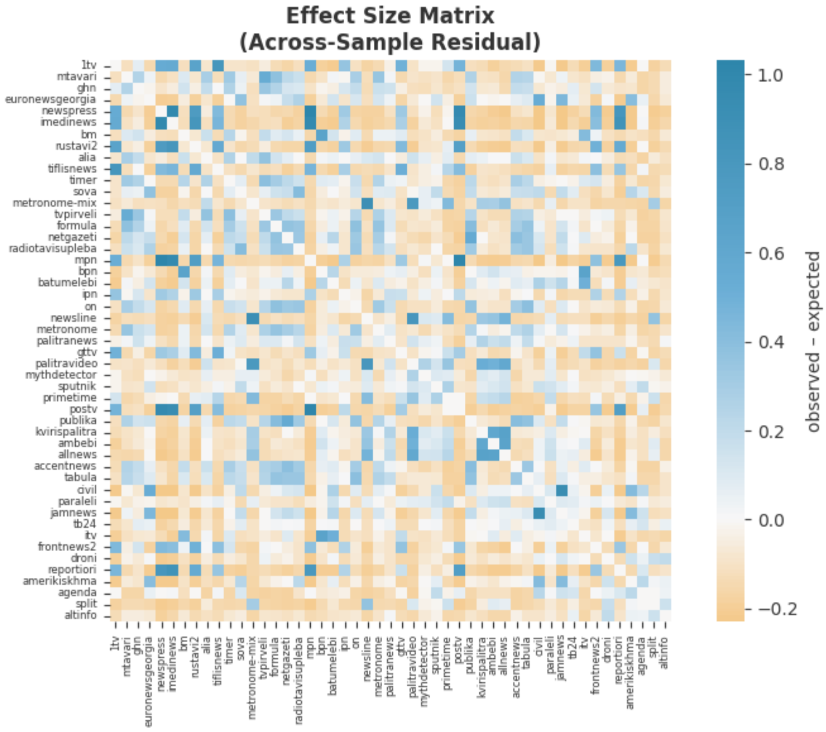

Figure A.10: Effect size matrix

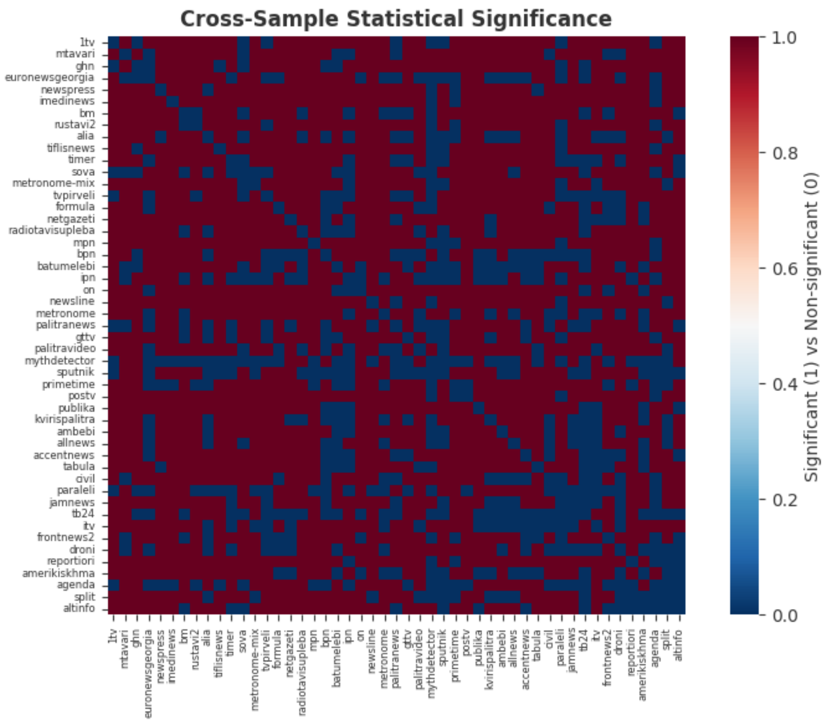

Figure A.11: Cross-sample statistical significance matrix

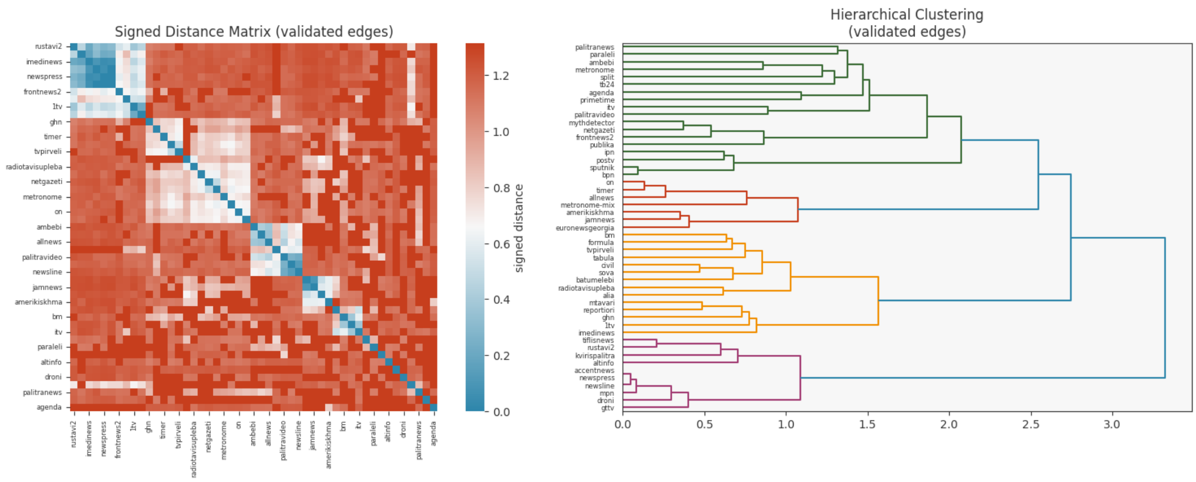

Figure A.12: (Left) Signed distance matrix using validated edges. (Right) Hierarchical clustering linkage tree.